COS 700 // RESEARCH POSTER

NLP-Based Anomaly Detection in 5G Network Logs

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1 Background & problem

Huge volume | **Heterogeneous** | **Unstructured**

Modern 5G networks generate large volumes of semi-structured log data that are difficult to analyse manually due to their scale and complexity. These logs contain critical information for system monitoring, fault diagnosis, and security anomaly detection, making automated analysis essential for reliable network operations.

Research problem: do these representations provide complementary evidence, and does performance depend on the classifier and prediction unit?

2 Aim & research questions

Aim: compare standalone and hybrid NLP representations for anomaly detection in 5G logs at entry and session levels.

- **RQ1** What challenges do 5G logs pose for NLP-based anomaly detection?
- **RQ2** How can TF-IDF, Word2Vec and a LogBERT-style model represent entries and sessions?
- **RQ3** How do Logistic Regression and Random Forest perform with each representation?
- **RQ4** Are hybrid models feasible, and do they improve detection?

Why compare? Representations preserve different signals, and simple methods can rival complex ones [25, 30].

3 Dataset

50,000	8,326	288	25%
synthetic log records	sessions of 5–7 events	message templates	abnormal in each log split

Table 1. Training, validation and test partitions

Partition	Log entries	Sessions
Train	35,000	5,834
Validation	7,500	1,246
Test	7,500	1,246

Group-based split: no scenario or session crosses partitions.

4 Methodology

Quantitative, supervised, comparative design.

Python-generated synthetic 5G core-network logs: 75% normal records and 25% abnormal records.

Clean & audit
lowercase; identifiers to placeholders; numbers to num_value; severity and technical terms kept

Train fits models → **Validation** picks thresholds → **Test** used once

Category 1: log entry
each message alone: TF-IDF, Word2Vec

Category 2: session
ordered events: TF-IDF, LogBERT-style

Five strategies × two classifiers
10 configurations per category, 20 in total

Figure 1. Experimental design and comparison workflow.

Table 2. Representation strategies

Strategy	Log-entry input	Session input
TF-IDF	Cleaned entry	Joined session text
Embedding	Mean Word2Vec	LogBERT-style vector
Concatenation	TF-IDF + Word2Vec	TF-IDF + contextual
Weighted	IDF-weighted vectors	IDF-weighted events
Decision fusion	Fused probabilities	Fused probabilities

Models & evaluation

- TF-IDF: word unigrams/bigrams; Word2Vec: 100-d vectors.
- LogBERT-style: two-layer masked-key Transformer.
- LR / RF (300 trees), training seed 42.
- Metrics: accuracy, precision, recall, F1 and FPR.

5 Results: F1-score

99.52% Log entry best F1
TF-IDF + LR: 0 false alarms, 18 missed

99.84% Session best F1
TF-IDF / concat. + RF tie
0 false alarms, 1 missed



Figure 2. Test F1-scores for log entries (top) and sessions (bottom).

Concat. = combined features; Weighted = IDF-weighted embedding; Fusion = combined probabilities.
★ Best F1 within each prediction unit (including ties).

6 Error analysis

Entry: TF-IDF + LR

Pred. normal	Pred. anomaly
5,625 TN	0 FP
18 FN	1,857 TP

Session: TF-IDF + RF

Pred. normal	Pred. anomaly
936 TN	0 FP
1 FN	309 TP

Figure 3. Test confusion matrices: entry TF-IDF + LR (left) and session TF-IDF + RF (right).

High recall can increase false alarms. Weighted LogBERT-style + LR raised 466 false alarms. Session fusion + RF detected all 310 anomalies, with 3 false alarms.

7 Prototype

5G Network Log Analysis

Flask web app

Python, scikit-learn, Gensim, PyTorch

7-step workflow

1 Upload | 2 Context | 3 Train | 4 Evaluate | 5 Compare | 6 Predict | 7 Dashboard

Figure 4. Seven-step prototype workflow.

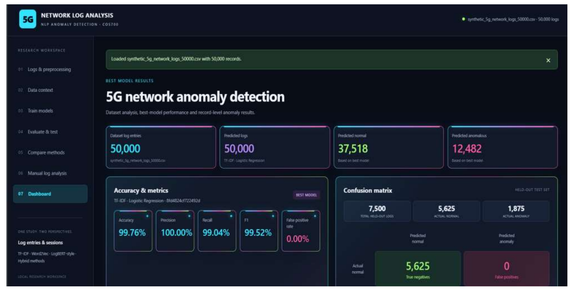


Figure 5. Prototype dashboard showing dataset predictions, model metrics and confusion counts.

8 Conclusions

- **RQ1** Volume, mixed fields and variable identifiers require cleaning.
- **RQ2** TF-IDF keeps exact words; Word2Vec pools them; LogBERT-style keeps event order.
- **RQ3** LR led for entries and RF for sessions. Classifier performance depends on the unit.
- **RQ4** Hybrids did not consistently improve on the best standalone model.

Limitations
Synthetic data with repeated templates shared by train and test; one training seed; a project-specific LogBERT-style model.

Future work
Real 5G logs, template-disjoint tests, repeated seeds with uncertainty estimates, and ablations of label-linked signals.

Take-home

TF-IDF led on this controlled corpus: best F1 99.52% for entries and 99.84% for sessions. Hybrid and LogBERT-style representations were feasible, but none consistently outperformed the strongest standalone model.

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Key references

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